

PHYS580 Lab 1

Assignment: We will perform the get-acquainted activities in (1) together as a hands-on exercise, while the remainder of the activities (2) through (5) is for you to perform and hand in a detailed report on at the beginning of the lab session next week. No need to include the activities in (1) in your report.

Lab 1 Activities:

1. This part is the hands-on introduction to basic tasks using computers and Pythia/Matlab... This task is worded with the assumption that you are using the lab computers. If that is not the case, adjust the activity accordingly.
 - a) Log in to your Purdue account and create a folder dedicated to PHYS580 and then a sub-folder for Lab 1 within it.
 - b) Start Pythia/Matlab (or another language/software of your choice), designating how and where your programs will reside and be called.
 - c) Learn to do simple plots of a one-variable function such as $f(x) = e^{-x/3} \sin(5x)$.
 - d) Learn to read input and display/print output. For example, create a text file containing 10 numbers (one in each line). Program to read the file and compute and output the sum of those numbers.
 - e) Say we have a function of (x,y,t) , e.g., $f(x,y,t) = \sin(2\pi x) \sin(2\pi y) \sin(2\pi t)$ for x,y in the range of $[0,1]$. Learn to visualize this as a movie as you vary t .
 - f) Go to the course home page on Brightspace and download the starter Pythia/Matlab program for Lab 1 to your Lab 1 area. Make copies before editing it to work on the rest of the assignment. (Or use them as guidance to write corresponding programs in your chosen other languages.)
2. Implement Euler approximation for the nuclear decay problem of Chap.1 in Pythia/Matlab (or in any language of your choice if you are familiar with it and do not wish to make use of the provided starter codes) for $t = 0$ to 5τ for a range of parameters, at least including $\Delta t/\tau = 0.05, 0.2, 0.8, 1$, and 1.5 . Plot the Euler approximation results together with the exact ones.
3. Calculate and analyze the deviations of your Euler results in (2) from the exact result in terms of their dependence on the step size $\Delta t/\tau$ (at fixed t) **and** on t (number n of steps used up to t) at fixed step size $\Delta t/\tau$, in the range $t = 0$ through 5τ . Include the graphic display of the cumulative deviations (*global* error) both in absolute terms and also as fractions of the exact values as a function of t .
4. Implement Runge-Kutta 2nd and 4th order approximations and do the same as in (2) and (3). How do they compare with the exact solution and with the Euler approximation results? Make the corresponding error analysis as you did in (3).
5. Based on your findings above, can you make any general observation on the approximation methods used here?